

Project

NextGen Smart Manufacturing Platform

1. Overview

The NextGen Smart Manufacturing Platform integrates IoT devices, edge computing, cloud analytics, AI-based predictive maintenance, and enterprise applications to provide real-time monitoring and optimization of manufacturing operations.

2. System Components

Industrial Sensors



Edge Gateway



IoT Message Broker (MQTT)



Data Processing Layer



Cloud Data Platform



AI Analytics Engine



Web Dashboard



3. Major Modules

- Sensor Management
 - Device Authentication
 - Data Collection
 - Event Processing
 - Predictive Analytics
 - Alert Management
 - Reporting
 - User Management
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4. Technology Stack

Layer	Technology
Sensors	Industrial IoT
Gateway	Edge Linux
Messaging	MQTT
Backend	Python / Java
Database	PostgreSQL
Analytics	AI Models
Dashboard	React
Cloud	Google Cloud Platform

5. Security Architecture

- OAuth2 Authentication
- TLS Encryption
- Role-Based Access Control

- Audit Logging
 - Data Encryption
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6. Scalability

Supports

- 100 Plants
 - 20,000 Devices
 - 5 Million Sensor Events/day
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7. Risks

- Network latency
- Device failures
- Data synchronization delays
- Cloud outage